$$A = \int_0^{\ln(2)} \left(e^x - e^{-x}\right) dx$$

$$A = \left[e^x - \frac{e^{-x}}{(-1)}\right]_0^{\ln(2)}$$

$$A = \left[e^{\ln(2)} + e^{-\ln(2)}\right] - \left[e^0 + e^0\right]$$

$$A = \left[2 + \frac{1}{2}\right] - \left[2\right]$$

$$A = \frac{5}{2} - 2 = \frac{1}{2}$$

$$\bar{x} = \frac{1}{A} \int_0^{\ln(2)} x\left(e^x - e^{-x}\right) dx$$

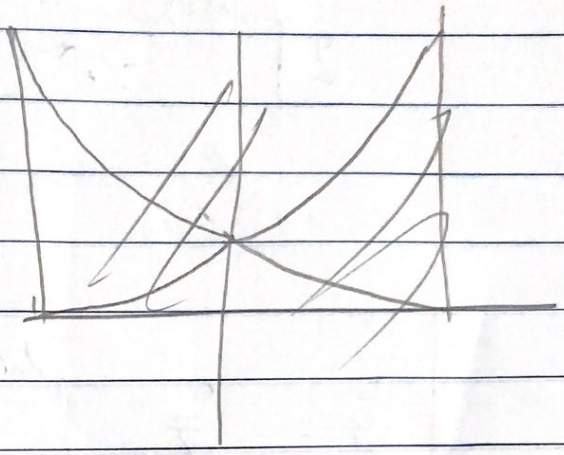
$$= 2 \int_0^{\ln(2)} xe^x \, dx - 2 \int_0^{\ln(2)} xe^{-x} \, dx$$

$$= 5 \ln(2) - 3$$

$$\bar{y} = \frac{1}{A} \int_0^{\ln(2)} \frac{1}{2} \left(\left(e^x\right)^2 - \left(e^{-x}\right)^2\right) dx$$

$$\bar{y} = 2\left(\frac{9}{16}\right) = \frac{9}{8}$$

$$(\bar{x}, \bar{y}) = \left(5\ln(2) - 3, \frac{9}{8}\right)$$



$y = x^2$

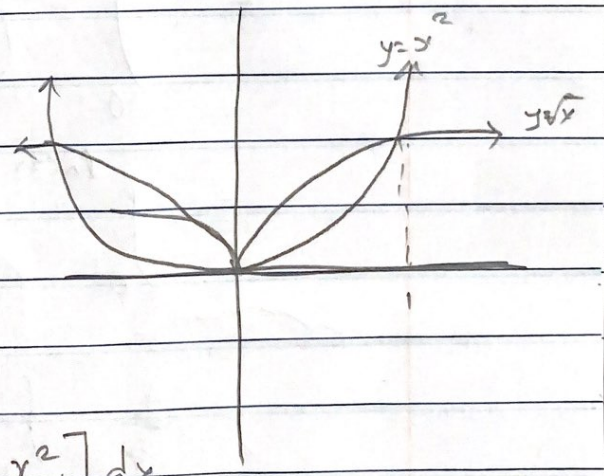
$y = \sqrt{x}$

y-axis

$$V = \int_a^b 2\pi x [f(x) - g(x)]\, dx$$

$$V = 2\pi^2 r^2 R$$

$$V = \pi r^2 \cdot 2\pi \bar{x}$$



$$V = \int_0^1 2\pi x \left[\sqrt{x} - x^2\right] dx$$

$$V = 2\pi \int_0^1 x^{3/2} - x^3\, dx$$

$$V = 2\pi \left[\frac{x^{5/2}}{5/2} - \frac{x^4}{4}\right]_0^1$$

$$V = 2\pi \left[\frac{2}{5} - \frac{1}{4}\right] = 2\pi \left[\frac{8-5}{20}\right] = \frac{3\pi}{10}$$

Notes

(1) $\bar{x} = 2\int_0^{\ln(2)} x e^{-x}\,dx = 1 - \ln(2)$

$2\int_0^{\ln(2)} x e^{x}\,dx = \left(2\ln(2) - 2\right) - \left(1 - \ln(2)\right)$

$= 3\ln(2) - 3$

$\bar{y} = \frac{\pi}{4}\int_0^{\ln(2)} \left((e^{x})^2 - (e^{-x})^2\right) dx$

$= \int_0^{\ln(2)} \left(e^{2x} - e^{-2x}\right) dx$

$= \int_0^{\ln(2)} \left(e^{x} + e^{-x}\right)\left(e^{x} - e^{-x}\right) dx$

$= \frac{\pi}{4}\int_2^{5/2} u\,du = \left[\frac{u^2}{2}\right]_2^{5/2} = \frac{25}{8} - 2$

$= 9/8$

05: $y = e^{x}$, $y = e^{-x}$, $x = 0$, and $x = \ln(2)$

